

Convolution, Duhamel's Principle, and Volterra Equations

A Self-Study Reference & Master Guide

Contents

Part I — Theory & Worked Examples	1
4.2 Convolution — Properties and Pitfalls	1
4.3 Duhamel's Principle — Response to Arbitrary Input	3
4.4 Volterra Integral Equations	5
Part II — Practice Set	8
Part III — Complete Solutions	9

Part I — Theory & Worked Examples

Convolution at a Glance: Core Sub-Topics

- **Algebraic Properties (4.2):** convolution as a true “product” in the time domain — commutative, associative, distributive, scalar-linear, and annihilated by zero — plus the pitfalls that separate it from ordinary multiplication.
- **Duhamel's Principle (4.3):** the impulse-response viewpoint. Once the response $h(t)$ to a unit impulse is known, the system's response to *any* input is the convolution $h * f$.
- **Volterra Integral Equations (4.4):** equations with *memory*, where the unknown $x(t)$ appears inside a convolution integral. The transform turns them into one-line algebra in s .

4.2 Convolution — Properties and Pitfalls

The convolution theorem tells us that a product of transforms $F(s)G(s)$ inverts to the convolution $(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$. Before we put this to work, it pays to treat convolution as an algebraic operation in its own right — one that behaves like multiplication in some ways, and dangerously *unlike* it in others.

Algebraic Properties

For piecewise-continuous, exponentially-bounded functions f, g, h and any constant c , convolution obeys the familiar algebra of a product:

- **Commutative:** $f * g = g * f$.
- **Associative:** $f * (g * h) = (f * g) * h$.
- **Distributive:** $f * (g + h) = f * g + f * h$.
- **Scalar-linear:** $(cf) * g = c(f * g)$.
- **Annihilated by zero:** $0 * f = 0$.

Each one mirrors a corresponding fact about the product $F(s)G(s)$ in the s -domain, which is why they hold.

No Multiplicative Identity; $f * f$ Can Be Negative

Convolution looks like multiplication, but two instincts carried over from ordinary products will burn you.

1. The constant 1 is *not* the identity: $f * 1 \neq f$. Convolving with 1 integrates:

$$(f * 1)(t) = \int_0^t f(\tau) \cdot 1 d\tau = \int_0^t f(\tau) d\tau.$$

For example $(\cos t) * 1 = \int_0^t \cos \tau d\tau = \sin t \neq \cos t$. The true convolution identity is the **Dirac delta**: by the sifting property,

$$(f * \delta)(t) = \int_0^t f(\tau) \delta(t - \tau) d\tau = f(t).$$

2. $f * f$ is not a square. Unlike $f(t)^2$, which is never negative, the convolution $f * f$ can dip below zero. For instance $(\sin t) * (\sin t) = \frac{1}{2}(\sin t - t \cos t)$, which is negative on a range of t . Never assume $f * f \geq 0$.

Worked Example 1: $(\cos t) * (\cos t)$

We evaluate the self-convolution of cosine directly from the definition.

Step 1 — Write the convolution integral.

$$(\cos * \cos)(t) = \int_0^t \cos(t - \tau) \cos \tau \, d\tau.$$

Step 2 — Apply the product-to-sum identity. Using $\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$ with $A = t - \tau$ and $B = \tau$, the difference is $A - B = t - 2\tau$ and the sum is $A + B = t$:

$$\cos(t - \tau) \cos \tau = \frac{1}{2} [\cos(t - 2\tau) + \cos t].$$

So the integral becomes

$$(\cos * \cos)(t) = \frac{1}{2} \int_0^t [\cos(t - 2\tau) + \cos t] \, d\tau.$$

Step 3 — Integrate term by term in τ . The variable of integration is τ , so the standalone $\cos t$ is a *constant* here — it pulls straight through the integral and contributes a factor of t :

$$\int_0^t \cos t \, d\tau = t \cos t.$$

For the other term, substitute $u = t - 2\tau$, $du = -2 \, d\tau$, so $\int \cos(t - 2\tau) \, d\tau = -\frac{1}{2} \sin(t - 2\tau)$. Evaluating from $\tau = 0$ to $\tau = t$:

$$\int_0^t \cos(t - 2\tau) \, d\tau = \left[-\frac{1}{2} \sin(t - 2\tau) \right]_0^t = -\frac{1}{2} \sin(-t) + \frac{1}{2} \sin(t) = \sin t.$$

Step 4 — Combine.

$$(\cos * \cos)(t) = \frac{1}{2} [\sin t + t \cos t].$$

The linearly growing factor $t \cos t$ is the fingerprint of resonance — a preview of Section 4.3.

4.3 Duhamel's Principle — Response to Arbitrary Input

Classical methods solve $x'' + \omega_0^2 x = f(t)$ one forcing function at a time. Duhamel's principle does it *once and for all*: it produces a single integral formula valid for *every* input f , built entirely from the system's response to a unit impulse.

The Killer Application — Four Steps

Solve $x'' + \omega_0^2 x = f(t)$ with rest initial conditions $x(0) = x'(0) = 0$, for an *arbitrary* forcing $f(t)$.

Step 1 — Transform. With zero initial conditions, $\mathcal{L}\{x''\} = s^2 X(s)$, so

$$s^2 X + \omega_0^2 X = F(s) \implies X(s) = \frac{F(s)}{s^2 + \omega_0^2}.$$

Step 2 — Identify the transfer function and impulse response. Factor off the input:

$$X(s) = H(s) F(s), \quad H(s) = \frac{1}{s^2 + \omega_0^2}, \quad h(t) = \mathcal{L}^{-1}\{H(s)\} = \frac{\sin(\omega_0 t)}{\omega_0}.$$

Here $h(t)$ is the *impulse response* — the output when $f = \delta(t)$, since $\mathcal{L}\{\delta\} = 1$.

Step 3 — Convolve. A product in s is a convolution in t , so

$$x(t) = (h * f)(t) = \int_0^t \frac{\sin(\omega_0(t - \tau))}{\omega_0} f(\tau) d\tau.$$

This single integral solves the ODE for *any* admissible f .

Step 4 — Resonance check. Feed the natural frequency back in: $f(t) = \cos(\omega_0 t)$. The convolution then produces a linearly growing amplitude,

$$x(t) = \frac{t \sin(\omega_0 t)}{2\omega_0},$$

the unbounded growth that signals resonance. The full computation is expanded below.

Expanding Step 4: Driving at the Natural Frequency

We carry out the convolution of Step 3 with $f(\tau) = \cos(\omega_0\tau)$ in full.

Step 1 — Set up the integral.

$$x(t) = \int_0^t \frac{\sin(\omega_0(t-\tau))}{\omega_0} \cos(\omega_0\tau) d\tau = \frac{1}{\omega_0} \int_0^t \sin(\omega_0(t-\tau)) \cos(\omega_0\tau) d\tau.$$

Step 2 — Product-to-sum. Use $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$ with $A = \omega_0(t-\tau)$ and $B = \omega_0\tau$. Then $A+B = \omega_0t$ and $A-B = \omega_0(t-2\tau)$:

$$x(t) = \frac{1}{2\omega_0} \int_0^t [\sin(\omega_0t) + \sin(\omega_0(t-2\tau))] d\tau.$$

Step 3 — Integrate in τ . The term $\sin(\omega_0t)$ is constant in τ , so it integrates to $t \sin(\omega_0t)$. The second term integrates to zero over $[0, t]$:

$$\int_0^t \sin(\omega_0(t-2\tau)) d\tau = \left[\frac{1}{2\omega_0} \cos(\omega_0(t-2\tau)) \right]_0^t = \frac{1}{2\omega_0} [\cos(-\omega_0t) - \cos(\omega_0t)] = 0,$$

because $\cos(-\theta) = \cos \theta$ makes the two boundary terms cancel.

Step 4 — Combine.

$$x(t) = \frac{1}{2\omega_0} t \sin(\omega_0t) = \frac{t \sin(\omega_0t)}{2\omega_0}.$$

The amplitude grows without bound as $t \rightarrow \infty$: this is resonance, falling out of the convolution automatically when the forcing frequency matches the natural frequency ω_0 .

Key Takeaway

Once the **impulse response** $h(t)$ is known, the system's response to *any* input is the convolution $x = h * f$. **The system is fully characterized by h** — no new solve is needed when the forcing changes.

4.4 Volterra Integral Equations

Differential equations describe systems whose future depends only on their *present* state. Many physical systems instead carry *memory*: the present response depends on the entire history of the input. Viscoelastic materials “remember” past deformation; age-structured populations depend on their full birth history; radiative transfer accumulates contributions along a path. These are modeled by **integral equations**, where the unknown sits inside a convolution. The Laplace transform dissolves the memory integral into ordinary algebra.

Solving by Algebra in s

A Volterra integral equation of convolution type has the form

$$x(t) = f(t) + \int_0^t g(t - \tau) x(\tau) d\tau.$$

The integral is exactly the convolution $(g*x)(t)$, so applying $\mathcal{L}$ and using $\mathcal{L}\{g*x\} = G(s)X(s)$:

$$X(s) = F(s) + G(s) X(s) \quad \implies \quad X(s)[1 - G(s)] = F(s) \quad \implies \quad \boxed{X(s) = \frac{F(s)}{1 - G(s)}}.$$

The unknown is isolated by simple division — the integral equation has become an algebraic one.

Worked Example 2: A Volterra Equation with a sinh Kernel

Solve $x(t) = e^{-t} + \int_0^t \sinh(t - \tau) x(\tau) d\tau$.

Step 1 — Transform both sides. The integral is the convolution $(\sinh * x)(t)$. With $\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$ and $\mathcal{L}\{\sinh t\} = \frac{1}{s^2-1}$:

$$X(s) = \frac{1}{s+1} + \frac{1}{s^2-1} X(s).$$

Step 2 — Collect the X terms. Move the X term to the left and factor it out:

$$X(s) - \frac{1}{s^2-1} X(s) = \frac{1}{s+1} \implies X(s) \left(1 - \frac{1}{s^2-1}\right) = \frac{1}{s+1}.$$

Step 3 — Common denominator inside the bracket.

$$1 - \frac{1}{s^2-1} = \frac{(s^2-1)-1}{s^2-1} = \frac{s^2-2}{s^2-1}, \quad \text{so} \quad X(s) \cdot \frac{s^2-2}{s^2-1} = \frac{1}{s+1}.$$

Step 4 — Multiply by the reciprocal and cancel. Solving for $X(s)$ and using $s^2-1 = (s-1)(s+1)$:

$$X(s) = \frac{1}{s+1} \cdot \frac{s^2-1}{s^2-2} = \frac{(s-1)(s+1)}{(s+1)(s^2-2)} = \frac{s-1}{s^2-2}.$$

Step 5 — Split into table-ready pieces. Decompose by separating the numerator:

$$X(s) = \frac{s-1}{s^2-2} = \frac{s}{s^2-2} - \frac{1}{s^2-2} = \frac{s}{s^2-2} - \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{s^2-2}.$$

Step 6 — Invert. Matching $\frac{s}{s^2-a^2} \rightarrow \cosh(at)$ and $\frac{a}{s^2-a^2} \rightarrow \sinh(at)$ with $a = \sqrt{2}$:

$$x(t) = \cosh(\sqrt{2}t) - \frac{1}{\sqrt{2}} \sinh(\sqrt{2}t).$$

Key Takeaway

A convolution-type integral equation transforms to $X(s) = \frac{F(s)}{1-G(s)}$ — the **memory integral becomes a single division**. Integral equations of physics-with-memory (viscoelasticity, age-structured populations, radiative transfer) yield to Laplace just as cleanly as differential equations.

Part II — Practice Set

Work each problem before turning to the complete solutions in Part III. Both reward setting up the integral or the s -domain algebra carefully before computing.

Problem 1: A Convolution Evaluation.

Evaluate the convolution $t * \sin t$ directly from the definition,

$$(t * \sin t)(t) = \int_0^t (t - \tau) \sin \tau \, d\tau,$$

and confirm your answer using the convolution theorem $\mathcal{L}\{t * \sin t\} = \mathcal{L}\{t\} \cdot \mathcal{L}\{\sin t\}$.

Problem 2: A Volterra Integral Equation.

Solve for $x(t)$:

$$x(t) = t + \int_0^t \sin(t - \tau) x(\tau) \, d\tau.$$

*Hint: the integral is a convolution $(\sin * x)(t)$; transform, isolate $X(s)$, and invert.*

Part III — Complete Solutions

Solution 1: $t * \sin t$.

We evaluate the integral directly, then verify in the s -domain.

Step 1 — Split the integral. Distribute $(t - \tau)$, remembering that t is constant with respect to the integration variable τ :

$$\int_0^t (t - \tau) \sin \tau \, d\tau = t \int_0^t \sin \tau \, d\tau - \int_0^t \tau \sin \tau \, d\tau.$$

Step 2 — First piece.

$$t \int_0^t \sin \tau \, d\tau = t [-\cos \tau]_0^t = t(1 - \cos t) = t - t \cos t.$$

Step 3 — Second piece by parts. Take $u = \tau$, $dv = \sin \tau \, d\tau$, so $du = d\tau$, $v = -\cos \tau$:

$$\int_0^t \tau \sin \tau \, d\tau = [-\tau \cos \tau]_0^t + \int_0^t \cos \tau \, d\tau = -t \cos t + \sin t.$$

Step 4 — Combine.

$$(t * \sin t)(t) = (t - t \cos t) - (-t \cos t + \sin t) = t - \sin t.$$

Step 5 — Verify via the convolution theorem.

$$\mathcal{L}\{t * \sin t\} = \frac{1}{s^2} \cdot \frac{1}{s^2 + 1} = \frac{1}{s^2(s^2 + 1)} = \frac{1}{s^2} - \frac{1}{s^2 + 1},$$

which inverts term by term to $t - \sin t$ — in agreement with the direct integration.

Solution 2: The Volterra Equation $x(t) = t + \int_0^t \sin(t - \tau)x(\tau) \, d\tau$.

The integral is the convolution $(\sin * x)(t)$, so we transform and solve for $X(s)$.

Step 1 — Transform. With $\mathcal{L}\{t\} = \frac{1}{s^2}$ and $\mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}$:

$$X(s) = \frac{1}{s^2} + \frac{1}{s^2 + 1} X(s).$$

Step 2 — Isolate $X(s)$. Collect terms and factor:

$$X(s) \left(1 - \frac{1}{s^2 + 1}\right) = \frac{1}{s^2}, \quad 1 - \frac{1}{s^2 + 1} = \frac{(s^2 + 1) - 1}{s^2 + 1} = \frac{s^2}{s^2 + 1}.$$

Step 3 — Multiply by the reciprocal.

$$X(s) \cdot \frac{s^2}{s^2 + 1} = \frac{1}{s^2} \implies X(s) = \frac{1}{s^2} \cdot \frac{s^2 + 1}{s^2} = \frac{s^2 + 1}{s^4}.$$

Step 4 — Split into table-ready pieces.

$$X(s) = \frac{s^2 + 1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}.$$

Step 5 — Invert. Using $\mathcal{L}^{-1}\{1/s^2\} = t$ and $\mathcal{L}^{-1}\{1/s^4\} = \frac{t^3}{3!} = \frac{t^3}{6}$:

$$x(t) = t + \frac{t^3}{6}.$$

Closing Takeaway

Both problems collapse to the same two moves: recognize the **convolution structure**, then trade it for a **product (or a division) in s** . Direct integration and the s -domain shortcut must always agree — a built-in check on every convolution computation.